DO NOW — RECALL

Lesson 064 · Thinking Concurrently: Independent Work

Name: _____ Class: _____ Date: _____

1. What is a process dependency?

2. What is throughput?

3. What is a race condition?

4. A rising-edge D-type flip-flop starts with Q=0. At a rising edge D=1, so Q becomes 1. Immediately afterwards D becomes 0. Explain why Q stays 1 through the following falling edge.

DO NOW — CHECK / OBJECTIVES

Correct and improve

Check page 1 in GREEN PEN. Write corrections in full.

1. Model answer

A task requires another task's result.

2. Model answer

Amount of completed work per unit time.

3. Model answer

Result depends on unsafe timing/order of shared operations.

4. Model answer

Only a rising edge samples D. Changing D and reaching a falling edge do not update Q, so it holds 1 until the next rising edge samples 0.

By the end, you can…

• Find tasks that do not depend on each other's intermediate results. Independent work can overlap.

• Distinguish concurrency (progress interleaved) from parallelism (truly simultaneous hardware execution).

• Evaluate scheduling overhead and shared-resource/race risks, and explain why dependencies limit possible speedup.

Vocabulary — know these terms

concurrency — Multiple tasks make progress in overlapping period.

parallelism — Actual simultaneous execution.

critical path — Longest dependency chain.

race condition — Incorrect timing-dependent shared state.

NEW LEARNING

Read, annotate and explain

Identify

Find tasks that do not depend on each other's intermediate results. Independent work can overlap.

Model

Distinguish concurrency (progress interleaved) from parallelism (truly simultaneous hardware execution).

Evaluate

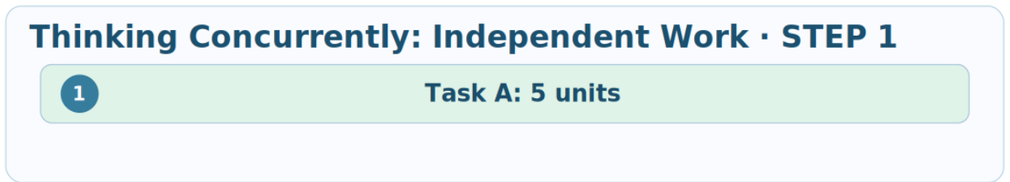
Evaluate scheduling overhead and shared-resource/race risks, and explain why dependencies limit possible speedup.

Explain one step in your own words, using the correct technical terms.

New Learning · original lesson diagrams

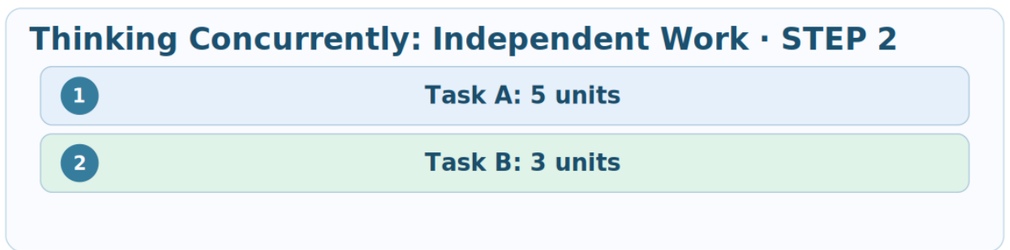
Diagrams appear after the related explanation. Annotate the original source visual.

New Learning



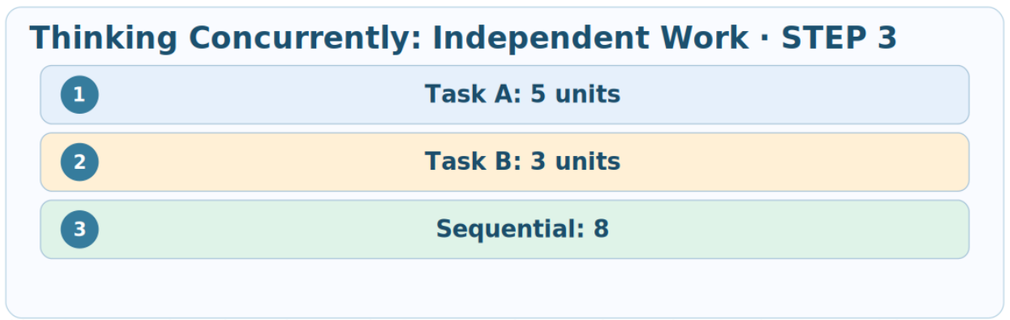
Thinking Concurrently: Independent Work · STEP 1: Task A: 5 units

New Learning



Thinking Concurrently: Independent Work · STEP 2: Task A: 5 units; Task B: 3 units

New Learning



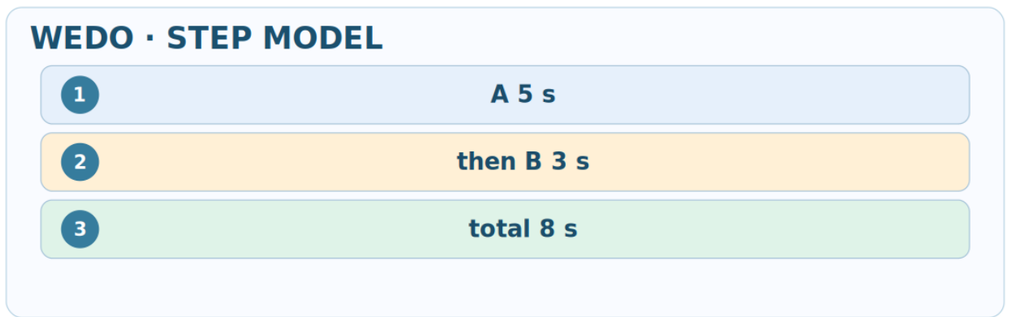
Thinking Concurrently: Independent Work · STEP 3: Task A: 5 units; Task B: 3 units; Sequential: 8

We Do source CHECK · original lesson diagrams

Diagrams appear after the related explanation. Annotate the original source visual.

Task B takes 3s but depends on Task A's output (A takes 5s). Can separate cores shorten ideal 8s chain?

Extra processors do not remove logical dependency.



WEDO · STEP MODEL: A 5 s; then B 3 s; total 8 s

WE DO — CHECK AND IMPROVE

Source-grounded model answers

CHECK only after completing the guided practice and Exit Ticket. Use GREEN PEN to improve your method and explain any error.

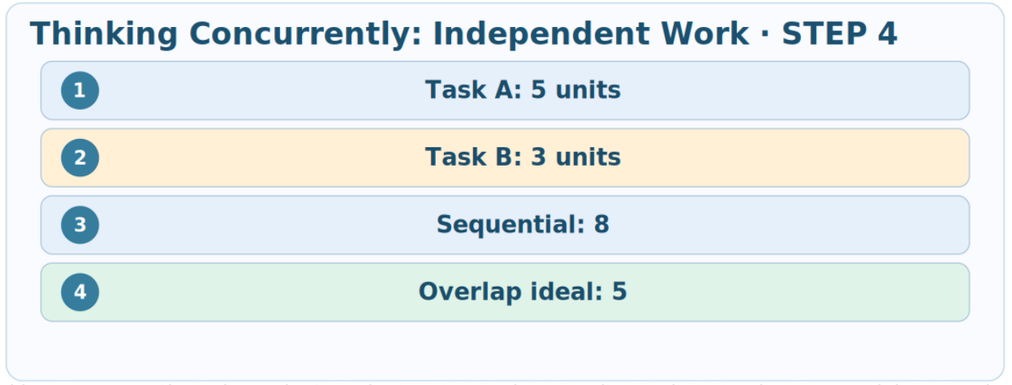
Guided 1 — CHECK

Explain the decision
Extra processors do not remove logical dependency.

New Learning · original lesson diagrams

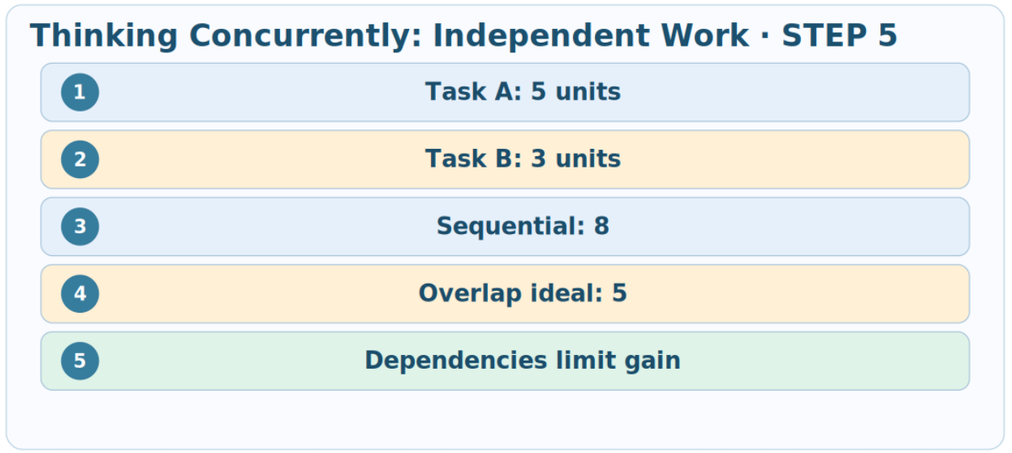
Diagrams appear after the related explanation. Annotate the original source visual.

New Learning



Thinking Concurrently: Independent Work · STEP 4: Task A: 5 units; Task B: 3 units; Sequential: 8; Overlap ideal: 5

New Learning



Thinking Concurrently: Independent Work · STEP 5: Task A: 5 units; Task B: 3 units; Sequential: 8; Overlap ideal: 5; Dependencies limit gain

I DO — WORKED EXAMPLE

Follow the worked method

Calculate minimum ideal time for independent image operations 6s and 4s on separate cores.

Explain the decision

Only independent tasks can safely overlap without waiting for each other.

EXIT TICKET / REFLECTION

Check your understanding

Complete the actual lesson Exit Ticket before checking.

Doubling the number of CPU cores always halves task completion time.

Circle: Confident / Getting there / Need help

Explain what you can now do independently.

GREEN PEN — Correct and explain

Choose one error. Write the correct method and explain precisely why your first response was wrong.

YOU DO — INDEPENDENT PRACTICE

Questions · Part 2

Answer independently. Show code, calculations or a justified explanation.

Question 3

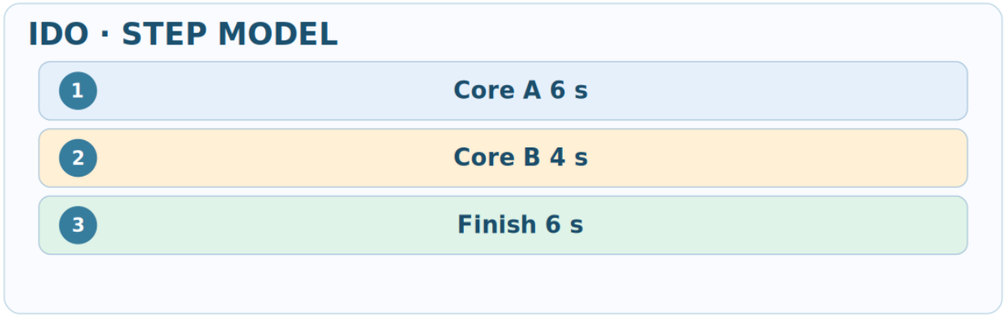
Two threads increment shared counter via read, add, write and occasionally lose updates. Explain race and one prevention. [6]

I Do · original lesson diagrams

Diagrams appear after the related explanation. Annotate the original source visual.

Calculate minimum ideal time for independent image operations 6s and 4s on separate cores.

Only independent tasks can safely overlap without waiting for each other.



IDO · STEP MODEL: Core A 6 s; Core B 4 s; Finish 6 s

WE DO — GUIDED PRACTICE

Try → Discuss → Improve

TRY: Attempt the guided questions in BLACK PEN. DISCUSS: Compare approaches with a partner, explain your reasoning and agree one improvement. IMPROVE: After the Exit Ticket, use the source CHECK model in GREEN PEN to correct and explain errors.

Guided 1 — TRY

Task B takes 3s but depends on Task A's output (A takes 5s). Can separate cores shorten ideal 8s chain?
Source code / input:
No: B cannot begin until A completes, so critical path is 5+3=8s.

YOU DO — INDEPENDENT PRACTICE

Questions · Part 1

Answer independently. Show code, calculations or a justified explanation.

Question 1

Independent jobs take 7s and 4s. Calculate sequential and ideal parallel completion times. [4]

Question 2

Give two potentially concurrent stages of a school app and two stages that must be sequential. [6]